DHOLE PATIL EDUCATION SOCIETY'S
DHOLE PATIL COLLEGE OF ENGINEERING

Department of First year Engineering

Course Code: ESC-102-ELE | **Subject: Basic Electrical Engineering** | **Academic Year – 2025-26**

Question Bank | **Semester –I & II**

Unit –II Electromagnetism		CO	BT	Marks
S.NO:	**Questions**			
Que.1	Define a) Magnetic flux, Flux density, Magnetic field intensity. b) Reluctance & its unit c) Permeance and its unit.	CO2	01	06
Que.2	Define Permeability and derive its expression	CO2	02	06
Que.3	State Faraday's law of Electromagnetic Induction.	CO2	01	06
Que.4	Explain with neat diagram Self Induced EMF and derive the expression for self inductance	CO2	02	06
Que.5	Compare Electric circuit and Magnetic circuit, clearing stating similar and dissimilar points.(Any eight points)	CO2	04	06
Que.6	A non magnetic ring has a mean diameter of 44.5cm and a cross sectional area of 12cm^2.It is uniformly wound with 500 turns. Calculate the Field strength and Total flux Produced in the ring by a current of 1A.	CO2	03	06
Que.7	A magnetic circuit consists of an iron ring of mean circumference 80 cm with a cross-sectional area 12cm2. A current of 2 A in the magnetizing coil of 200 turns produces a total flux of 1.2mWb in the iron. Calculate: i) flux density in the iron ii) absolute and relative permeability of iron iii) reluctance of the circuit	CO2	03	06
Que.8	Define Mutually Induced EMF with a neat sketch And derive expression for Mutual Inductance	CO2	02	06
Que.9	A coil of 500 turns is wound on an iron ring of circular cross sectional area 3 cm^2 and mean diameter 20 cm. The magnetic flux produced in the ring is 0.5 mWb when the coil carries 2 A current. Determine the relative permeability of the material.	CO2	03	06
Que.10	A coil is wound with 300 turns on a steel ring of relative permeability of 900. It has a mean circumference of 40 cm and cross sectional area of 5cm^2. If the coil has resistance of 100 ohm and it is connected to 250 V DC supply. Calculate: (a) Current in the coil (b) MMF (b) Field Strength (c) Total flux (d) Reluctance	CO2	05	06

Que.11	Explain Series Magnetic Circuit and derive the expression with air-gap.	CO2	01	06
Que.12	State the factor on which self inductance and mutual inductance depends?	CO2	01	06
Que.13	Derive expression for energy stored in inductor per unit volume	CO2	05	06
Que.14	Derive expression for coefficient of coupling.	CO2	02	06
Que.15	A closed magnetic circuit is composed of two sections in series. Section A has a length of 40 cm and cross sectional area of 10mm2. Section B has a length of 50 cm and cross sectional area of 14mm2. Both the sections are made up of same material having permeability of 650. A coil with 400 turns is wound over one of the section. Find the current required in the coil so as to develop a flux density of 1.4 Tesla in section B.	CO2	03	06
Que.16	A coil of 500 turns and resistance 20 Ohm is wound uniformly on an iron ring of mean circumference 50 cm and cross-sectional area 4 cm^2. It is connected to a 24 V d. c. supply. Under these conditions, the relative permeability of iron is 800. Calculate the values of (i) Magnetomotive force of the coil (ii) Magnetising force (iii) Total flux in iron (iv) Reluctance of the ring	CO2	03	06